

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 50%

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50, 100, 150, 200]$$

Compute the transformed values.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency	
0	2
1	2
2	4

Intensity Frequency

3 8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

[10 20 30 20 30 40 30 40 50]

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

[10 10 10, 10 20 10, 10 10 10]

Compute the output at the center.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$F = [100, 80, 40, 20]$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

[200]

Apply threshold $T = 100$. Assign values (0 or 1).

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$\begin{bmatrix} -1 & -2 & -1, & 0 & 0 & 0, & 1 & 2 & 1 \end{bmatrix}$

to:

[10 10 10, 20 20 20, 30 30 30]

Compute the gradient at the center.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:

[1 0 2]

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

[1 0 5]

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

11. Power-Law (Gamma) Transformation

A grayscale image is enhanced using power-law transformation.

(a) Interpret the role of gamma correction in image enhancement.

(b) Justify the use of transformation $s = cr^\gamma$, where $\gamma = 0.5$, $c = 1$.

(c) Apply the transformation step-by-step to pixel values:

$r = [25, 100, 225]$.

(d) Compute the transformed values accurately.

(e) Interpret the results in terms of brightness enhancement.

12. Contrast Stretching

An image has intensity values in the range [50, 150].

(a) Interpret the need for contrast stretching.

(b) Justify linear transformation to range [0, 255].

(c) Apply transformation step-by-step for values:

$r = [50, 100, 150]$.

(d) Compute the output values accurately.

(e) Interpret improvement in contrast.

13. Histogram Equalization (4-bit Image)

A 4-bit image has the following histogram:

Intensity Frequency

0	4
1	6
2	10
3	6

- (a) Interpret the need for equalization.
- (b) Justify use of CDF.
- (c) Compute cumulative distribution.
- (d) Compute new intensity levels.
- (e) Interpret contrast enhancement.

14. Median Filtering

A noisy image contains salt-and-pepper noise.

- (a) Interpret noise characteristics.
- (b) Justify median filter usage.
- (c) Apply filter to:

10 200 10,200 50 200,10 200 10

- (d) Compute center value.
- (e) Interpret noise removal effectiveness.

15. Gaussian Smoothing

An image is smoothed using Gaussian filter.

- (a) Interpret smoothing objective.
- (b) Justify Gaussian filter over averaging.
- (c) Apply filter:

[1 2 1, 2 4 2, 1 2 1]

to matrix:

[10 20 10,20 40 20,10 20 10]

- (d) Compute center pixel.
- (e) Interpret smoothing effect.

16. High-Pass Filtering

An image requires sharpening.

- (a) Interpret need for high-pass filtering.
- (b) Justify filter selection.

(c) Apply mask:

$[-1 \ -1 \ -1, -1 \ 8 \ -1, -1 \ -1 \ -1]$

(d) Compute center value.

(e) Interpret edge enhancement.

17. Frequency Domain High-Pass

Given frequency components:

$$F = [10, 20, 80, 100]$$

(a) Interpret high-frequency importance.

(b) Justify high-pass filtering.

(c) Apply filtering (remove low frequencies).

(d) Compute output.

(e) Interpret result.

18. Adaptive Thresholding

Pixel values: [60, 80, 120, 140]

(a) Interpret segmentation challenge.

(b) Justify adaptive thresholding.

(c) Apply local threshold (mean = 100).

(d) Compute binary output.

(e) Interpret result.

19. Edge Detection (Vertical Sobel)

Apply vertical Sobel:

$$[-1 \ 0 \ 1, -2 \ 0 \ 2, -1 \ 0 \ 1]$$

Matrix:

$$[10 \ 20 \ 30, 10 \ 20 \ 30, 10 \ 20 \ 30]$$

(a) Interpret edge detection objective.

(b) Justify operator choice.

(c) Apply step-by-step.

(d) Compute center value.

(e) Interpret edge direction.

20. Prewitt Edge Detection

(a) Interpret Prewitt operator purpose.

(b) Justify use over Sobel.

(c) Apply kernel:

$[-1 \ 0 \ 1, -1 \ 0 \ 1, -1 \ 0 \ 1]$

(d) Compute output.

(e) Interpret result.

21. Region Splitting

Pixel set: [40, 45, 100, 105]

(a) Interpret segmentation problem.

(b) Justify splitting method.

(c) Apply threshold 80.

(d) Compute regions.

(e) Interpret separation.

22. Region Merging

Regions: [45, 50], [52, 55]

(a) Interpret merging criteria.

(b) Justify merging approach.

(c) Apply threshold = 5.

(d) Compute merged region.

(e) Interpret result.

23. Edge Linking

Edges detected: [1, 0, 1, 0, 1]

(a) Interpret problem of discontinuity.

(b) Justify linking method.

(c) Apply linking step.

(d) Compute result.

(e) Interpret continuity.

24. Low-Pass Filtering (Frequency)

Given $F = [200, 150, 50, 20]$

(a) Interpret smoothing requirement.

(b) Justify LPF selection.

(c) Apply filter.

- (d) Compute output.
- (e) Interpret effect.

25. Laplacian Sharpening (Variant)

Mask:

$$\begin{array}{ccc} 0 & -1 & 0, \\ -1 & 5 & -1, \\ 0 & -1 & 0 \end{array}$$

- (a) Interpret sharpening need.
- (b) Justify mask.
- (c) Apply to matrix.
- (d) Compute center.
- (e) Interpret result.

26. Histogram Matching

- (a) Interpret need for matching.
- (b) Justify method.
- (c) Map values [10, 50, 100].
- (d) Compute new values.
- (e) Interpret.

27. Noise Reduction Comparison

- (a) Interpret noise type.
- (b) Compare averaging vs median.
- (c) Apply both.
- (d) Compute outputs.
- (e) Interpret differences.

28. Combined Filtering

- (a) Interpret need for smoothing + sharpening.
- (b) Justify sequence.
- (c) Apply filters stepwise.
- (d) Compute output.
- (e) Interpret improvement.

29. Multi-Threshold Segmentation

Pixels: [20, 60, 120, 180]

- (a) Interpret multi-level segmentation.
- (b) Justify thresholds (50,100).
- (c) Apply segmentation.
- (d) Compute classes.
- (e) Interpret.

30. Boundary Detection

- (a) Interpret boundary extraction.
- (b) Justify method.
- (c) Apply edge + linking.
- (d) Compute result.
- (e) Interpret boundaries.

Code of Conduct:

I S. Yuva Guru Mani Varma /192425005 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Yuva Guru Mani Varma

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2-AT-1

S-Yuva Guru Mani Varma

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Analytical Problem Solving

ITA0504 - Computer Vision

1) Grey level Transformation

a) Improves image contrast by changing pixel intensity values

b) Process:

* Formula $\Rightarrow S: 255 - r$

$\Rightarrow 0 \rightarrow 255$

$\Rightarrow 50 \rightarrow 205$

$\Rightarrow 100 \rightarrow 155$

$\Rightarrow 150 \rightarrow 105$

$\Rightarrow 255 \rightarrow 0$

Ans: 255, 205, 155, 105, 0

2) Log Transformation

a) Brightens dark pixels.

b) Process:

* Formula: $S = \lceil \ln(1+r) \rceil$

* $\ln(1) = 0$

* $\ln(2) = 0.693$

* $\ln(4) = 1.386$

* $\ln(8) = 2.079$

Ans: 0, 0.693, 1.386, 2.079

3) Histogram Equalization

a) Improves image contrast

b) Process:

* Total Pixels = 16

* CDF = 2, 8, 8, 16

* New intensity = $(CDF/16) \times 7$

Ans: 1, 2, 4, 7

6) Smoothing Filter

a) Removes Filter noise.

b) Process:

* Add all 9 pixels

* Divide by 9.

* Average = 47

Ans: Center Pixel = 47

5) Sharpening

a) Enhances Edges

b) Process:

* Apply Laplacian mask.

* Multiply mask with neighbourhood.

* Add values

Ans: Center pixel = 80

6) Frequency Domain Filter

a) LPF smooths image; HPF sharpens image.

b) Process:

* Keep only first two frequency components

* Remove remaining frequencies

Ans: Smooth image obtained.

7) Thresholding

a) Separates foreground and background

b) Process: Threshold = 100

* $60 < 100 \rightarrow 0$

* $90 < 100 \rightarrow 0$

* $120 \geq 100 \rightarrow 1$; Ans: 0011

Sobel Edge Detection

a) Detect Edges

- b) Process: * Apply Sobel kernel
* Multiply corresponding values.
* Add results

Ans: Gradient = 80.

Region Growing

a) Groups similar pixels

b) Process: Seed = 50

* $|48 - 50| = 2 \checkmark$

* $|52 - 50| = 2 \checkmark$

* $|60 - 50| = 10 \times$

Ans: Region = 48, 50, 52, 54.

Smoothing + Thresholding

a) Removes noise before segmentation

b) Process: $\gamma < 1$ brightness image.

(c, d): $\sqrt{0} = 0$, $\sqrt{25} = 5$

$\sqrt{100} = 10$

$\sqrt{225} = 15$

Ans: 0, 5, 10, 15.

Gamma Transformation

a) Controls Brightness

b) Process: Mean = 48

Threshold = 50

$48 < 50 \rightarrow 0$

Ans: 00 11

12) Contrast Stretching

a) Improves Contrast

(b, c, d): Process: Formula $\Rightarrow s = \frac{(r - 50)}{100} \times 255$

* $50 \rightarrow 0$

* $75 \rightarrow 64$

* $100 \rightarrow 128$

* $150 \rightarrow 255$

Ans: 0, 64, 128, 255

13) Histogram Equalization

a) Improves image contrast.

b) CDF preserves intensity order.

c) Process: * CDF $\Rightarrow 4, 10, 20, 26$

d) Process: * New level = $(\text{CDF}/26) \times 15$
* New values = 2, 6, 12, 15

e) Contrast is improved.

14) Median Filtering

a) Removes salt and pepper noise

b) Preserves edges better than averaging.

c) Process: Arrange values: 20, 24, 25, 16, 27, 28, 29, 30, 255

d) Median = 27

e) Noise is removed efficiently.

15) Gaussian Smoothing

a) Reduces image noise

b) Gives more weight to nearby pixels.

c) Process: * Apply Gaussian kernel
* Divide result by 16.

Center pixel = 27

Smooth pixel with edges.

High-Pass Filtering

- a) Enhances Edges.
- b) Highlights fine details
- c) Apply high-pass mask.
- d) Center value = 80
- e) Sharper image.

1) Frequency Domain High-Pass

- a) High frequencies contain edge information.
- b) Removes smooth background.
- c) Remove low frequencies.
- d) Output: High frequency components only.
- e) Sharper result.

18) Adaptive Thresholding

- a) Handles uneven lighting , b) Uses local threshold.
- c) Threshold = 100 , d) Output: 0, 0, 1, 1 , e) Better Segmentation.

19) Vertical Sobel

- a) Detects vertical Edges , b) Better noise suppression , c) Apply Sobel Kernel
- d) Centre Value = 80 , e) Strong vertical edges.

20) Prewitt Edge Detection.

- a) Detects Image Gradients , b) Simpler than sobel , c) Apply Prewitt Kernel
- d) Output = 80 , e) Indicates vertical edge.

21) Region Splitting

- a) Split regions with large intensity differences, b) Creates homogeneous
- c) Threshold = 80, d) Regions: {40, 50}, {100, 105}, e) Correct

22) Region Merging

- a) Merge similar neighbouring regions, b) Based on intensity diff.
- c) Difference = 6 > 5, d) No Merging, e) Regions remain separate

23) Edge linking:

- a) Connect broken edges, b) Improves boundary continuity, c) Process

Original: 10101

After linking: 11111, d) Result = 11111, e) Continuous edges obtained

24) Low-Pass Filtering.

- a) Removes high-frequency noise, b) Reserves low frequencies,
- c) Process: * Remove high freq, d) Output = Low-freq components

25) Laplacian Sharpening

- a) Enhances details, b) Detects edges in all directions, Process:
- * Apply Laplacian mask, d) Center pixel = 80, e) Sharper image.

26) Histogram Matching.

- a) Matches reference histogram.
- b) Gives consistent appearance.
- c) Process: 10 → 20
50 → 80
100 → 160, d) Output: 20, 80, 160, e) Better brightness.

Noise Reduction Comparison.

a) Salt - and - pepper noise.

b) Process:

$$\text{Average} = \text{Sum} / 9$$

$$\text{Median} = \text{Middle Value.}$$

c) Average = 49

d) Median = 24

e) Median filter is better.

8) Combined Filtering

a) Removes noise then sharpens.

b) Prevents noise amplification.

c) Process:

Average : Sharpen.

50 $\rightarrow$ Smooth $\rightarrow$ Sharpen.

d) Final output = 50

e) Better image quality.

29) Multi-Thread Segmentation

a) Divides image into multiple ~~regions~~

b) Uses thresholds 50 to 100.

c) Process: $20 < 50 \rightarrow \text{Class 1}$

$60 \rightarrow \text{Class 2}$

$120 \rightarrow \text{Class 3}$

d) Classes : 1, 2, 3, 3.

e) Better object separation.

30) Boundary Detection.

a) Detects objects boundaries.

b) Uses edge ~~edge~~ and linking.

c) Process:

Detect edges $\rightarrow$ Link ~~broken~~ edges.

d) Output : Continuous Boundary.

e) Improves object recognition.